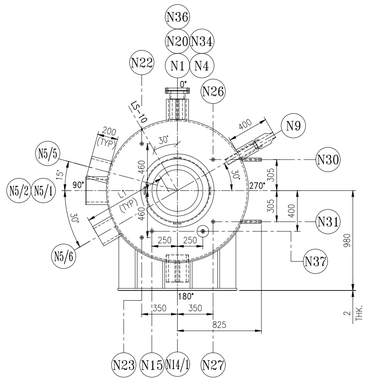
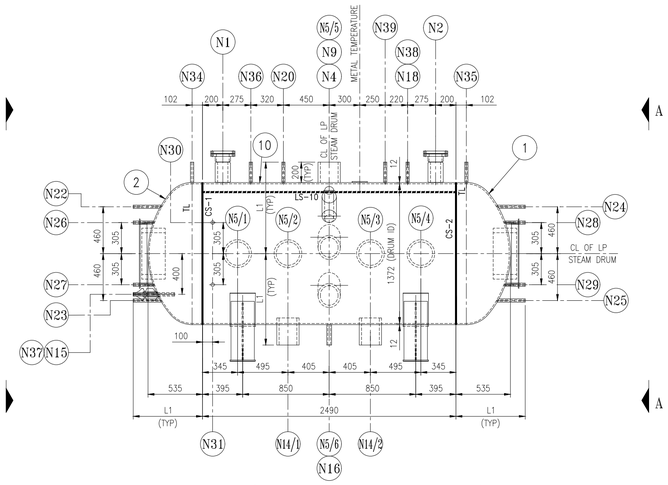
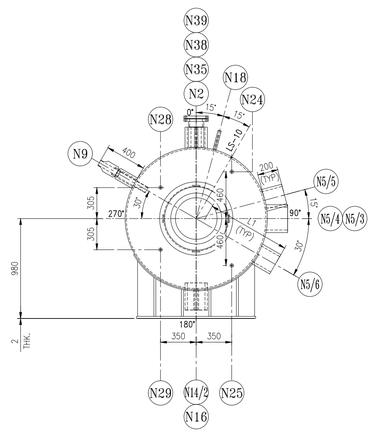
PROJECT ID	JB1164
DOCUMENT ID	JB1164-50010100-007-0001



VIEW – BB

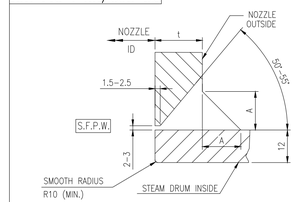


ELEVATION (VIEWED FROM FRONT OF BOILER)
ARRANGEMENT OF STEAM DRUM

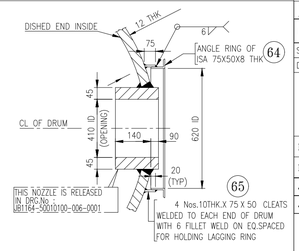


VIEW – AA

IBR CHAPTER – XII
IBR FIG : XII/85E B2



(MARK Nos. N1, N2)
WELD DETAIL OF NOZZLE TO STEAM DRUM



DETAIL OF MANHOLE OPENING

DESIGN DATA

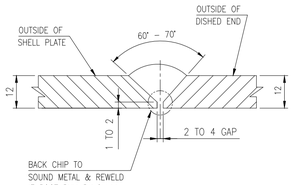
CODE OF CONSTRUCTION	I.B.R. (1950) & IT'S LATEST AMMENDMENTS UPTO OCTOBER 2019 (SEE NOTE : 01)
MAXIMUM WORKING PRESSURE	6.0 Kg/Sq. cm (g)
MAXIMUM WORKING TEMPERATURE	164° C
WORKING METAL TEMPERATURE	164° C
HYDRAULIC TEST PRESSURE	9.0 Kg/Sq. cm (g) AT SHOP & AS PER IBR AT SITE
RADIOGRAPHY OF LONG.&CIRCUM.SEAM	100%
STRESS RELIEVING	YES & REFER HEAT TREATMENT CYCLE TABLE
EVAPORATION OF BOILER (M.C.R.)	8.65 TPH
TOTAL HEATING SURFACE AREA OF BOILER	5295 Sq.m (APPROX.)
HEATING SURFACE AREA OF ECONOMISER (NON STEAMING)	NIL
HEATING SURFACE AREA OF DRUM	NIL (OUT SIDE GAS PATH)
STEAM PRESSURE AT MSSV OUTLET	4.0 Kg/Sq. cm (g)
STEAM TEMPERATURE AT MSSV OUTLET	195°C
WEIGHT OF DRUM	2991.05 Kgs. (Δ1)
INSPECTION	DIRECTOR OF BOILERS, UTTAR PRADESH/Lloyds (SEE NOTE : 02)

MATERIAL SPECIFICATIONS :

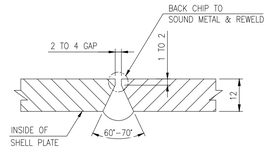
DESCRIPTION	SIZE	SPECIFICATION	MIN. TENSILITY (Kg/ Sq.mm)	MIN.CALC. THK. (mm)	REMARKS
SHELL BELTS	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	3.69	FOR STEAM DRUM
DISHED ENDS	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	3.73	FOR STEAM DRUM

LIST OF REFERENCE DRAWINGS

DRAWING DOCUMENT ID	DESCRIPTION
JB1164-50010100-006-0001	ARRANGEMENT & DETAIL OF CIRCULAR MANHOLE ON LP STEAM DRUM FOR PH
JB1164-50010100-009-0001	DETAILS OF LP STEAM DRUM NOZZLES FOR PH

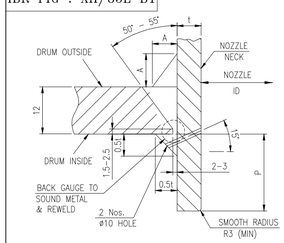


WELD DETAIL OF CLOSING SEAM FOR STEAM DRUM
(SEAM MARK Nos. CS-1 & CS-2)



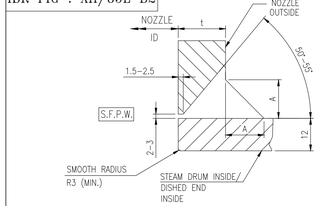
WELD DETAIL OF LONGITUDINAL SEAM
FOR STEAM DRUM
(SEAM MARK Nos. LS-10)

IBR CHAPTER – XII
IBR FIG : XII/85E B1



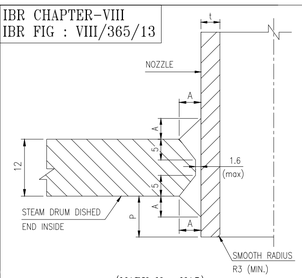
(FOR MARK Nos. N14/1 & N14/2)
WELD DETAIL OF NOZZLE TO DRUM

IBR CHAPTER – XII
IBR FIG : XII/85E B2

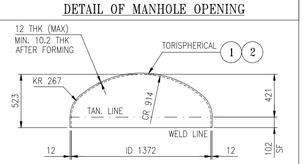


(MARK Nos. N4, N5/1 TO N5/6, N9, N16, N18, N20, N22, N23, N24, N25, N26, N27, N28, N29, N30, N31 N34, N35, N36, N38 & N39)
WELD DETAIL OF NOZZLE TO STEAM DRUM & DISHED END

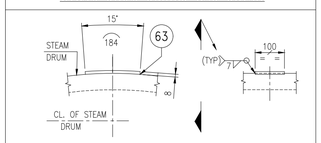
IBR CHAPTER–VIII
IBR FIG : VIII/365/13



(MARK No. N15)
WELD DETAIL OF NOZZLE TO DISHED END



STEAM DRUM DISHED END DETAIL



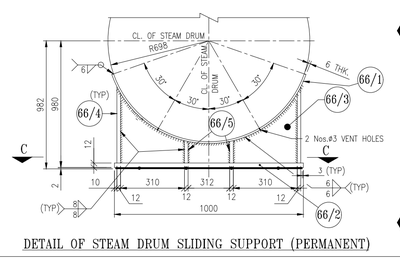
DETAIL OF METAL TEMPERATION

PWHT CYCLE :- (FOR LP STEAM DRUM)

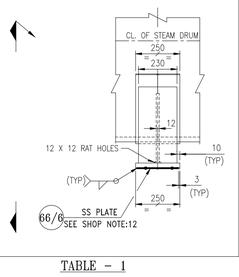
A)	LOADING TEMP. (MAX.)	= 300°C
B)	RATE OF HEATING ABOVE 300°C	= 120°C/Hr.
C)	SOAKING TEMP.	= 610°±10°C
D)	SOAKING TIME (MIN.)	= 30 min.
E)	RATE OF COOLING UP TO 300°C	= 120°C/Hr.
F)	UNLOADING TEMP.	= 300°C
G)	COOLING AFTER UNLOADING	= COOLING IN STILL AIR

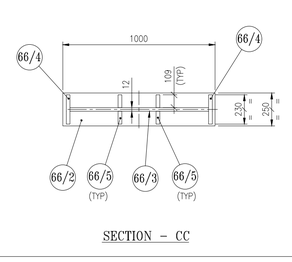
Δ1	--	FLATS & BRACKETS ON DRUM	--	--	1	28.27 JB1164-50010100-013-0002
	--	FLATS & STUDS INSIDE DRUM	--	--	1	85.26 JB1164-50010100-012-0001
Δ1	--	DET. OF DRUM NOZZLE	--	--	1	582.85 JB1164-50010100-009-0001
	--	A&O OF MAN HOLE DOOR	--	--	2	516.08 JB1164-50010100-006-0001
	66/6	SS PLATE	SA 240 Gr. 304	PL. 2 THK – 994 X 244	2	7.62 –
	66/5	SUPPORT PLATE	IS 2062 Gr. E250 A	PL. 12 THK – 285 X 109	8	23.44 –
	66/4	SUPPORT PLATE	IS 2062 Gr. E250 A	PL. 12 THK – 463 X 230	4	40.12 –
	66/3	SUPPORT PLATE	IS 2062 Gr. E250 A	PL. 12 THK – 956 X 451	2	81.22 AS DRAWN
	66/2	BASE PLATE	IS 2062 Gr. E250 A	PL. 12 THK – 1000 X 250	2	47.10 –
	66/1	SADDLE PAD	IS 2062 Gr. E250 A	PL. 6 THK – 1468 X 250	2	34.58 –
	65	CLEAT FOR RING	IS 2062 Gr. E250 A	PL. 10 THK – 75 X 50	8	2.32 –
	64	LAGGING RING	IS 2062 Gr. E250 A	ISA 75 X 50 X 8THK–2105 Lg.	2	31.15 –
	63	PAD	SA 516 Gr.70	PL. 8 THK – 100 X 185	1	1.16 FOR METAL TEMPERATURE
	10	SHELL	SA 516 Gr.70	PL. 12 THK–4550 X 2500	1	990.44 (1071.5 Kg.–MAT.FORECAST WT.)
	2	DISHED END	SA 516 Gr.70	PL. 12 THK– 1940 X 1940	1	259.57 (354.53 Kg.–MAT.FORECAST WT.)
	1	DISHED END	SA 516 Gr.70	PL. 12 THK– 1940 X 1940	1	259.87 (354.53 Kg.–MAT.FORECAST WT.)
SDM001		STEAM DRUM WITH ATTACHMENTS		P B215 1	(2991.05) Δ1	–

ITEM CODE	P.NO.	PART DESCRIPTION / ITEM DESCRIPTION	SPECIFICATION	SIZE	L.TYPE	L.GROUP	L.QTY	P.QTY	P.WT(Kg) I.WT (Kg) (TOTAL)	REMARK

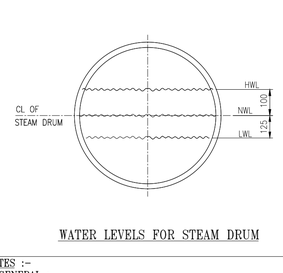


DETAIL OF STEAM DRUM SLIDING SUPPORT (PERMANENT)

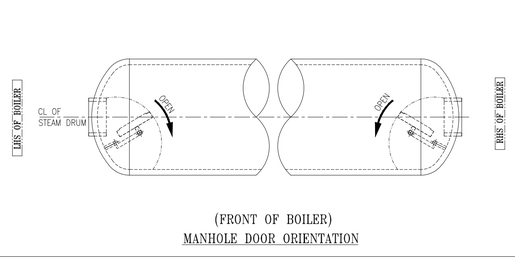




SECTION – CC



WATER LEVELS FOR STEAM DRUM



(FRONT OF BOILER)
MANHOLE DOOR ORIENTATION

TABLE – 1

	S.Nos.	Nozzle Nos.	Nozzle Descriptions	Location of Nozzle	No. of Nozzle	Nozzle Size (OD) mm / (D) mm	Schedule of Nozzle (t)	Nozzle Thickness (t) mm	Connecting Pipe Thickness (mm)	Projection from C.L of Drum (L) mm From Dished End Inside (L1) mm	Nozzle C.L to Drum Vertical Axis (A1) mm	Nozzle C.L to Drum Horizontal Axis (A2) mm	Calculated Reinforcing Pad Length (R) mm	Finished Length of Nozzle (N) mm	M/C Bore for end Preparation (B) mm	Weld Dim. (K) mm	Nozzle Inside Projection (p) mm
	1	N1	SAFETY VALVE NOZZLE–1	STEAM DRUM SHELL	1	168	–	14	–	–	–	–	198	198	80	12	0
Δ1	2	N2	SAFETY VALVE NOZZLE–2	STEAM DRUM SHELL	1	168	–	14	–	–	–	–	198	198	80	12	0
	3	N4	LP SUPER HEATER STEAM STUB	STEAM DRUM SHELL	1	219.1	SCH.160	23.01	SCH.20	898	–	–	253	203	206.40	12	0
	4	N5/1 TO N5/4	LP EVAPORATOR RISER STUB – I	STEAM DRUM SHELL	4	273.0	SCH.XXS	25.4	SCH.20	898	–	–	256	206	260.30	12	0
	5	N5/5 & N5/6	LP EVAPORATOR RISER STUB – II	STEAM DRUM SHELL	2	219.1	SCH.160	23.01	SCH.20	898	–	–	253	203	206.4	12	0
	6	N9	FEED INLET THERMAL SLEEVE	STEAM DRUM SHELL	1	114.3	SCH.XXS	17.12	REF.009–0001	1098	–	–	REF.DRG : –009–0001				
	7	N14/1 & N14/2	FEEDER NOZZLE	STEAM DRUM SHELL	2	219.1	SCH.160	23.01	SCH.20	898	–	–	322	272	206.40	12	60
	8	N15	CBD NOZZLE	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	680	250	400	499	449	26.64	9	150
	9	N16	IBD NOZZLE	STEAM DRUM SHELL	1	48.3	SCH.XXS	10.15	SCH.40	898	–	–	247	197	40.94	10	0
	10	N18	AIR VENT NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	898	–	–	247	197	26.64	9	0
	11	N20	NITROGEN BLANKETING NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	898	–	–	247	197	26.64	9	0
	12	N22	LEVEL GAUGE NOZZLE (STEAM)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	680	350	460	412	362	26.64	9	0
	13	N23	LEVEL GAUGE NOZZLE (WATER)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	680	350	460	412	362	26.64	9	0
	14	N24	LEVEL GAUGE NOZZLE (STEAM)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	680	350	460	412	362	26.64	9	0
	15	N25	LEVEL GAUGE NOZZLE (WATER)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	680	350	460	412	362	26.64	9	0
	16	N26	LEVEL TRANSMITTER (STEAM)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	680	350	305	330	280	26.64	9	0
	17	N27	LEVEL TRANSMITTER NOZZLE (WATER)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	680	350	305	330	280	26.64	9	0
	18	N28	LEVEL TRANSMITTER (STEAM)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	680	350	305	330	280	26.64	9	0
	19	N29	LEVEL TRANSMITTER NOZZLE (WATER)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	680	350	305	330	280	26.64	9	0
	20	N30	LEVEL TRANSMITTER (STEAM)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	825	–	–	256	206	26.64	9	0
	21	N31	LEVEL TRANSMITTER NOZZLE (WATER)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	825	–	–	256	206	26.64	9	0
	22	N34	PRESSURE GAUGE NOZZLE (LOCAL)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	898	–	–	250	200	26.64	9	0
	23	N35	REMOTE PRESSURE GAUGE NOZZLE (FLOOR)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	898	–	–	250	200	26.64	9	0
	24	N36	PRESSURE TRANSMITTER NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	898	–	–	247	197	26.64	9	0
	25	N37	HP DOSING NOZZLE WITH THERMAL SLEEVE	STEAM DRUM SHELL	1	115	–	32.0	REF.009–0001	680	–	–	REF.009–0001				150
	26	N38	PRESSURE TRANSMITTER NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	898	–	–	247	197	26.64	9	0
	27	N39	SPARE NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	–	898	–	–	247	197	–	9	0

NOTES :–

I) GENERAL :

01. THE ZERO DATE OF THIS CONTRACT IS 22.10.2019 HENCE IBR AMMENDMENTS UPTO OCTOBER 2019 HAVE BEEN TAKEN CARE OF WHILE DESIGNING THE EQUIPMENT.
02. INSPECTION BY DIRECTOR OF BOILERS, UTTAR PRADESH/Lloyds CBB Authorisation, form XVIA See Regulation 4C(2). BECAUSE MANUFACTURING OF THIS EQUIPMENT SHALL WILL BE DONE AT ATV PROJECT INDIA LIMITED, MATHURA, UTTAR PRADESH.
03. ALL DIMENSIONS ARE IN 'mm' UNLESS OTHERWISE SPECIFIED.
04. ALL FILLET WELDS ARE 6mm UNLESS OTHERWISE SPECIFIED.
05. FOR DIMENSIONAL TOLERANCE REQUIREMENTS PLEASE REFER DOC. NO. GEN–50990100–DSM–002 AND ITS ASSOCIATED DRAWINGS.
06. CIRCUMFERENTIAL DIMENSION (ON OUTSIDE TRAMMELS ON DRUM) LP STEAM DRUM – 4385.7 mm
07. LEGEND :

H.W.L	: DENOTES HIGH WATER LEVEL.	ID	: DENOTES INSIDE DIAMETER.
N.W.L	: DENOTES NORMAL WATER LEVEL.	SF	: DENOTES STRAIGHT FACE.
L.W.L	: DENOTES LOW WATER LEVEL.	CR	: DENOTES CROWN RADIUS.
CS–1&CS–2	: DENOTES CLOSING SEAM.	KR	: DENOTES KNUCKLE RADIUS.
LS–10	: LONGITUDINAL SEAM.	S.F.P.W	: SET ON FULL PENETRATION WELD
TL	: TANGENTIAL LINE.		
LP	: LOW PRESSURE.		

II) SHOP NOTES :

01. ALL CIRCUMFERENTIAL DIMENSIONS ARE MEASURED OUTSIDE OF DRUMS
02. MANHOLE CIRCULAR TYPE SIZE ø410 FOR DRUMS REF. DRG. No. JB1164–50010100–006–0001
03. A MAX. 4^{0} mm FILLET WELD TO BE USED FOR WELDING OF CLEAT BRACKETS AS PER DRG.No : 011–0001 TO 014–0001 AFTER STRESS RELIVING.
04. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH THE FOLLOWING DRGS.
 i. FOR NOZZLE & STUBS DETAILS REFER DRG.No : JB1164–50010100–009–0001
 ii. FOR EXTERNAL AND INTERNAL BAFFLES & PIPE REFER DRG.No : 010–0001 TO 015–0001.
05. ALL WELDING TO BE DONE BY IBR QUALIFIED WELDER USING IBR WELDING PROCEDURES
06. FOR BOILER DRUMS HAVING SHELL THICKNESS 100mm AND ABOVE FOLLOWING ADDITIONAL TESTING SHALL BE CARRIED OUT:
 i. ULTRASONIC EXAMINATION OF ALL BUTT WELD BEFORE AND AFTER PWHT.
 ii. MAGNETIC PARTICLE EXAMINATION OF NOZZLES/STUBS TO SHELL WELDING.
08. ALL THE PADS WELDED ALL–ROUND ON PRESSURE PART SHALL BE PROVIDED WITH 2 Nos 1/8" NPT(F) TELL–TALE HOLES AT 180° APART ON THE PAD TO RELEASE ENTRAPPED AIR/GAS.
09. TOLERANCE FOR FORMED HEADS:–
 THE INNER SURFACE OF A TORISPHERICAL SHALL NOT DEVIATE OUTSIDE OF THE SPECIFIED SHAPE BY MORE THAN $1\frac{1}{2}$ % OF NOMINAL INSIDE DIAMETER & INSIDE THE SPECIFIED SHAPE BY MORE THAN $\frac{5}{8}$ % OF NOMINAL INSIDE DIAMETER.
10. SUITABLE STIFFENER ARRANGEMENT SHALL BE PROVIDED AT SHOP FOR NOZZLES AT DISHED END TO AVOID DAMAGE DURING TRANSPORTATION.
11. MANHOLE DOORS ARE TO BE PLACED INSIDE THE DRUM BEFORE ASSEMBLING DRUM ENDS FOR FINAL CIRCUMFERENTIAL WELD AS PER ORIENTATION DETAIL.
12. 2mm THK. SS PLATE SHALL BE PROTECTED USING WOODEN BLANKS DURING TRANSPORTATION.
13. PRIOR TO HYDRAULIC TESTING FOLLOWING SHALL BE CARRIED OUT.
 a) ALL WELD SPATTER, WELD STUBS, SCALE, DIRT ETC SHALL BE REMOVED FROM DRUM.
14. EQUIPMENT SHALL BE THOROUGHLY DRIED/CLEANED AFTER HYDRO TESTING.
15. HYDRO TESTING TO BE CARRIED OUT WITH GASKET THAT IS THE SAME AS THE SPECIFIED SERVICE GASKET.
16. HYDROTEST WATER SHALL NOT CONTAIN MORE THAN 10ppm SUSPENDED SOLIDS.
17. ALL INSIDE EDGES OF NOZZLES AND CONNECTIONS SHALL BE GROUND SMOOTH TO 3 MM RADIUS.
18. STRAY ARCING SHOULD BE AVOIDED, IF ANY STRAY ARCING NOTED SHALL BE DRESSED, MPI & UT CHECKED.
19. FOLLOW APPROVED AND APPLICABLE WPS FOR SELECTION OF WELDING CONSUMABLES.
20. WELD SURFACES SHALL BE FREE FROM CRACKS, UNDERCUTS, UNNECESSARY GRINDING, ABRUPT RIDGES AND VALLEYS.
21. VISUAL INSPECTION OF WELD SHALL BE CARRIED OUT AND NOTED BEFORE OFFERING FOR ANY NON DESTRUCTIVE EXAMINATION.
22. HYDRO TEST WATER QUALITY SHALL BE MAINTAINED WITHIN LIMITS SPECIFIED IN DOCUMENT NO.– BLS–PPA–HYD–01–S1.
23. DRUM TO BE STAMPED AS FOLLOWS :–

ITEM CODE–SDM001–LP STEAM DRUM
ATV PROJECTS INDIA LTD., MATHURA (U.P)
(A/C ISGEC HEAVY ENGINEERING LTD.)
WORKS NO – /WT NO. 6665–LP
YEAR OF MAKE–
TESTED TO –9.0 Kg/Sq. cm (g)
TESTED ON –
MAXIMUM WORKING PRESSURE – 6.0 Kg/Sq. cm (g)
COMPETENT PERSON'S / INSPECTING AUTHORITY'S OFFICIAL STAMP
FINAL DATE OF EXAMINATION –

PAINTING NOTES:–

01. FOR PAINTING NOTES REFER SPECIFICATION NO:JB1164–50570900–PSH–0001.

BOILER MAKERS NUMBER I.W.T. No. : 6665–LP

I.W.T. No. : 6665–LP

CONSULTANT	--
CLIENT	M/s. ULTRATECH NATHDWARA CEMENT LIMITED, PINDWARA,SIROHI, RAJASTHAN.

DO NOT SCALE IF IN DOUBT PLEASE ASK

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FOR APPROVAL	YES	FOR INFORMATION	NO	FOR PRODUCTION	YES	FOR ERECTION	YES

RESPONSIBLE DEPT.	DRAWN BY	CHECKED BY	APPROVED BY	DATE
MECHANICAL	V.KAVASKAR	R.G.AMIR/R.KUMARAN	R.ARUNKUMAR	28.12.2019

TOTAL WEIGHT IN kg. 2991.05 Δ1

PROJECT ID	ELEMENT ID	I.W.T. NO.
JB1164	50010100	6665–LP

YEAR : 2019 SHEET SIZE : A1 SCALE :1 : 30

ISGEC BOILERS (formerly Isgec John Thompson)
Isgec Heavy Engineering Ltd.
Noida 201 301 (U.P.) India

SERVICE	WHR BOILER FOR PRE HEATER PH–2 (C –STRING)
TITLE	ARRANGEMENT & DETAILS OF LP STEAM DRUM FOR PH
DRAWING DOCUMENT ID	JB1164–50010100–007–0001
REVISION	01

REVN.	ZONE	DESCRIPTION	REVD. BY	CHKD. BY	APPD. BY	DATE
Δ1	≃	1. HOLD HAS BEEN REMOVED FOR SAFETY VALVE NOZZLE Nos. N1 & N2 AND CORRESPONDINGLY ITS WELD DETAIL, TABLE – 1 HAS BEEN UPDATED.	V.KAVASKAR	R.G.AMIR	R.ARUNKUMAR	12.02.2020

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